

PUSH DOWN AUTOMATA AND CONTEXT FREE LANGUAGE

(2 MARKS)

1. GATE CSE 2026 Set 2

MCQ (More than One Correct Answer) Marks: +2 / -0

Let $\Sigma = \{a, b, c, d\}$ and let $L = \{a^i b^j c^k d^l \mid i, j, k, l \geq 0\}$.

Which of the following constraints ensure(s) that the language L is context-free?

- (A) $i + k = j + l$
- (B) $i = k$ and $j = l$
- (C) $i = l$ and $j = k$
- (D) $i + j = k + l$

2. GATE CSE 2026 Set 1

MCQ (More than One Correct Answer) Marks: +2 / -0

Consider the following context-free grammar G :

$$\begin{aligned} S &\rightarrow abaABAbba \\ A &\rightarrow aaBBAb \mid bBabaa \\ B &\rightarrow aBb \mid ab \end{aligned}$$

In the above grammar, S is the start symbol, a and b are terminal symbols, and A and B are non-terminal symbols.

Let $L(G)$ be the language generated by the grammar G . For a string $s \in L(G)$, let $n_1(s)$ be the number of a's in s and $n_2(s)$ be the number of b's in s .

Which of the following statements is/are true?

- (A) There is a string $s \in L(G)$ such that $n_1(s) < n_2(s)$
- (B) For every string $s \in L(G)$, $n_1(s) \geq n_2(s)$
- (C) There is a string $s \in L(G)$ such that $n_1(s) > 2n_2(s)$
- (D) For every string $s \in L(G)$, $n_1(s) \leq 2n_2(s)$

3. GATE CSE 2025 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0

Consider the following two languages over the alphabet $\{a, b, c\}$, where m and n are natural numbers.

$$\begin{aligned} L_1 &= \{a^m b^m c^{m+n} \mid m, n \geq 1\} \\ L_2 &= \{a^m b^n c^{m+n} \mid m, n \geq 1\} \end{aligned}$$

Which ONE of the following statements is CORRECT?

- (A) Both L_1 and L_2 are context-free languages.
- (B) L_1 is a context-free language but L_2 is not a context-free language.
- (C) L_1 is not a context-free language but L_2 is a context-free language.
- (D) Neither L_1 nor L_2 are context-free languages.

4. GATE CSE 2024 Set 2

MCQ (More than One Correct Answer) Marks: +2 / -0

Consider a context-free grammar G with the following 3 rules.

$$S \rightarrow aS, \quad S \rightarrow aSbS, \quad S \rightarrow c$$

Let $w \in L(G)$.

Let $n_a(w)$, $n_b(w)$, $n_c(w)$ denote the number of times a , b , c occur in w , respectively. Which of the following statements is/are TRUE?

- (A) $n_a(w) > n_b(w)$
- (B) $n_a(w) > n_c(w) - 2$
- (C) $n_c(w) = n_b(w) + 1$
- (D) $n_c(w) = n_b(w) * 2$

5. GATE CSE 2024 Set 1

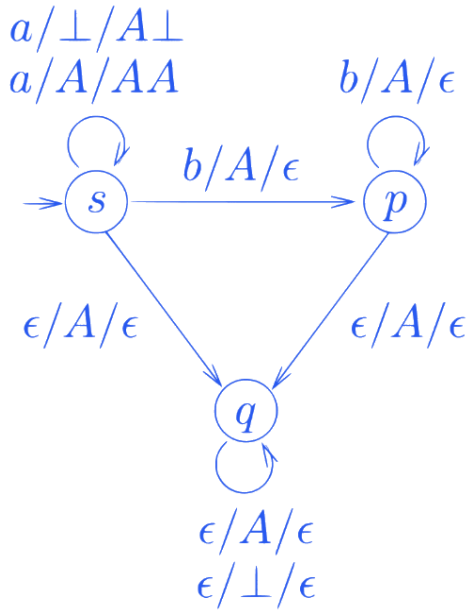
Numerical Marks: +2 / -0

Let $G = (V, \Sigma, S, P)$ be a context-free grammar in Chomsky Normal Form with $\Sigma = \{a, b, c\}$ and V containing 10 variable symbols including the start symbol S . The string $w = a^{30}b^{30}c^{30}$ is derivable from S . The number of steps (application of rules) in the derivation $S \Rightarrow^* w$ is _____

6. GATE CSE 2023

MCQ (Single Correct Answer) Marks: +2 / -0.67

Consider the pushdown automation (PDA) P below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{\perp, A\}$, and has three states $\{s, p, q\}$, with s being the start state. A transition from state u to state v , labelled $c/X/\gamma$, where c is an input symbol or ϵ , X is a stack symbol, and γ is a string of stack symbols, represents the fact that in state u , the PDA can read c from the input, with X on the top of its stack, pop X from the stack, push in the string γ on the stack, and go to state v . In the initial configuration, the stack has only the symbol $\perp$ in it. The PDA accepts by empty stack.



Which one of the following options correctly describes the language accepted by P?

- (A) $\{a^m b^n \mid 1 \leq m \text{ and } n < m\}$
- (B) $\{a^m b^n \mid 0 \leq n \leq m\}$
- (C) $\{a^m b^n \mid 0 \leq m \text{ and } 0 \leq n\}$
- (D) $\{a^m \mid 0 \leq m\} \cup \{b^n \mid 0 \leq n\}$

7. GATE CSE 2023

MCQ (Single Correct Answer) Marks: +2 / -0.67

Consider the context-free grammar G below

$$\begin{aligned} S &\rightarrow aSb \mid X \\ X &\rightarrow aX \mid Xb \mid a \mid b, \end{aligned}$$

where S and X are non-terminals, and a and b are terminal symbols. The starting non-terminal is S .

Which one of the following statements is CORRECT?

- (A) The language generated by G is $(a + b)^*$
- (B) The language generated by G is $a^*(a + b)b^*$
- (C) The language generated by G is $a^*b^*(a + b)$
- (D) The language generated by G is not a regular language

8. GATE CSE 2022

MCQ (More than One Correct Answer) Marks: +2 / -0

Consider the following languages:

$$L_1 = \{a^n w a^n \mid w \in \{a, b\}^*\}$$

$$L_2 = \{w x w^R \mid w, x \in \{a, b\}^*, |w|, |x| > 0\}$$

Note that w^R is the reversal of the string w . Which of the following is/are TRUE?

- (A) L_1 and L_2 are regular.
- (B) L_1 and L_2 are context-free
- (C) L_1 is regular and L_2 is context-free.
- (D) L_1 and L_2 are context-free but not regular.

9. GATE CSE 2022

MCQ (More than One Correct Answer) Marks: +2 / -0

Consider the following languages:

$$L_1 = \{w w \mid w \in \{a, b\}^*\}$$

$$L_2 = \{a^n b^n c^m \mid m, n \geq 0\}$$

$$L_3 = \{a^m b^n c^n \mid m, n \geq 0\}$$

Which of the following statements is/are FALSE?

- (A) L_1 is not context-free but L_2 and L_3 are deterministic context-free.
- (B) Neither L_1 nor L_2 is context-free.
- (C) L_2 , L_3 and $L_2 \cap L_3$ all are context-free.
- (D) Neither L_1 nor its complement is context-free.

10. GATE CSE 2021 Set 2

MCQ (More than One Correct Answer) Marks: +2 / -0

For a string w , we define w^R to be the reverse of w . For example, if $w = 01101$ then $w^R = 10110$.

Which of the following languages is/are context-free?

- (A) $\{x w x^R \mid w, x \in \{0, 1\}^*\}$
- (B) $\{w w^R x x^R \mid w, x \in \{0, 1\}^*\}$
- (C) $\{x w x^R \mid w, x \in \{0, 1\}^*\}$
- (D) $\{w x x^R w^R \mid w, x \in \{0, 1\}^*\}$

11. GATE CSE 2021 Set 1

Numerical Marks: +2 / -0

In a pushdown automaton $P = (Q, \Sigma, \Gamma, \delta, q_0, F)$, a transition of the form,



where $p, q \in Q$, $a \in \Sigma \cup \{\epsilon\}$, and $X, Y \in \Gamma \cup \{\epsilon\}$, represents $(q, Y) \in \delta(p, a, X)$.

Consider the following pushdown automaton over the input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{\#, A\}$.



The number of strings of length 100 accepted by the above pushdown automaton is _____

12. GATE CSE 2020

MCQ (Single Correct Answer) Marks: +2 / -0.67

Consider the following languages.

$$L_1 = \{wxyx \mid w, x, y \in (0 + 1)^+\}$$

$$L_2 = \{xy \mid x, y \in (a + b)^*, |x| = |y|, x \neq y\}$$

Which one of the following is TRUE?

- (A) L_1 is regular and L_2 is context-free.
- (B) L_1 is context-free but not regular and L_2 is context-free.
- (C) Neither L_1 nor L_2 is context-free.
- (D) L_1 is context-free but L_2 is not context-free.